IPCC Tier 2 Livestock GHG Uncertainty Calculator

Technical Methodology: Equations, Monte Carlo Procedure, and Uncertainty Reporting

CIAT / CGIAR Alliance — ClimateActionNetZero

Version 1.0

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Contents

1 Introduction

1.1 Purpose

The **IPCC Tier 2 Livestock GHG Uncertainty Calculator** is a web-based tool that enables national inventory compilers to quantify the uncertainty in cattle greenhouse gas (GHG) emission estimates produced using the IPCC Tier 2 methodology. The tool implements the Monte Carlo (Approach 2) uncertainty framework described in IPCC (2006) Volume 1, Chapter 3, applied to the full emission equation chain of IPCC (2006) Volume 4, Chapter 10 and Chapter 11 for pasture N_2O.

Under the Paris Agreement Enhanced Transparency Framework (ETF), national inventory reports are expected to include quantitative uncertainty estimates alongside emission totals. This tool automates that process, from parameter sampling to the formatted IPCC Table 3.3 output, for inventories of GHG emissions from one or more sub-categories of cattle.

1.2 Scope

The tool covers six emission sources for cattle (or buffalo):

1. **Enteric fermentation CH_4** — methane produced during enteric fermentation in the rumen
2. **Manure management CH_4** — methane from anaerobic decomposition of manure in storage systems
3. **Manure management N_2O (direct)** — direct nitrous oxide from nitrogen in managed manure
4. **Manure management N_2O (indirect)** — indirect N_2O via atmospheric deposition and leaching from managed manure
5. **Pasture/range/paddock N_2O (direct)** — direct N_2O from dung and urine deposited on pasture
6. **Pasture/range/paddock N_2O (indirect)** — indirect N_2O from dung and urine on pasture

The tool supports multiple cattle sub-categories (e.g. dairy cows, beef bulls, heifers, growing males, calves) within a single inventory run, with results aggregated to national totals and to per-cattle-type subtotals (dairy, non-dairy), and with a per-sub-category breakdown that users can use to represent production systems, regions, or animal classes.

1.3 IPCC Tier Classification

The tool implements **Tier 2** methods throughout the emission chain. Tier 2 enteric fermentation outputs (GE, DE, CP) feed the manure CH_4 and N_2O calculations, so the whole chain is classified as Tier 2. The uncertainty analysis itself follows the **Approach 2** (Monte Carlo) framework, which is the more accurate of the two IPCC uncertainty methods and is recommended when non-linear equation chains are involved.

2 Input Parameters

All parameters are entered via an Excel template (one row per parameter per cattle sub-category) and loaded into the tool. Each parameter requires a central value, an uncertainty specification (distribution type and bounds or percentage), and an optional IPCC reference.

Table 1 lists all parameters used in the calculation chain, their units, the IPCC equation in which they appear, the default value pre-loaded in the tool (sourced from the IPCC Guidelines), and an indication of potentially suitable probability distributions that can be applied in uncertainty analysis. Where the 2006 Guidelines and the 2019 Refinement give different defaults, both are shown.

Table 1: Input parameters for the IPCC Tier 2 cattle GHG calculation chain — including IPCC defaults and source citations.

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`N`	Number of animals (population)	head	*n/a*	Eq. 10.1 (Tier 1 totals); national activity data	Normal
`BW`	Mean live body weight	kg/head	275	Eq. 10.3; Annex 10A.2 (Africa cattle)	Normal
`Cfi`	Coefficient for NE_m	MJ/kg$^{0.75}$/d	0.386 (lact.) / 0.322 (non-lact.) / 0.370 (bulls)	Eq. 10.3; Table 10.4 (Updated)	Constant
`Tw`	Mean winter temperature	°C	20 (inert)	Eq. 10.2 (cold-climate Cfi adjustment)	Constant
`Ca`	Activity coefficient	—	0.00 (stall) / 0.17 (pasture flat) / 0.36 (large/hilly)	Eq. 10.4; Table 10.5 (Updated)	Constant
`WG`	Mean daily weight gain	kg/head/d	0 (non-growing default)	Eq. 10.6	Normal
`MW`	Mature body weight	kg/head	300	Eq. 10.6; Annex 10A.2	Normal
`C`	Sex/condition coefficient for NE_g	—	0.8 (female) / 1.0 (castrate) / 1.2 (bull)	Eq. 10.6	Constant

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Table 1 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`Milk`	Daily milk yield	kg/head/d	3.5 (Africa, IPCC 2019R)	Eq. 10.8; Annex 10A.1 / 10A.2	Normal/Lognormal
`MilkFat`	Milk fat content	%	4.3 (Africa, IPCC 2019R)	Eq. 10.8; Annex 10A.1	Normal
`pct_pregnant`	Fraction of females pregnant / yr	—	0.60	2019 Refinement Vol.4 Ch.10 §10.5	Beta
`MilkPR`	Milk protein content	%	3.3 ($\approx 1.9 + 0.4 \times$ %Fat)	Eq. 10.33 (uses 6.38 milk-protein-to-N); Annex 10A.1–10A.3	Normal
`WorkHours`	Daily working hours	h/head/d	0 (no work)	Eq. 10.11	Constant
`Cp`	Pregnancy coefficient	—	0.10 (cattle)	Eq. 10.13; Table 10.7 (Updated)	Constant
`DE_pct`	Digestible energy (% of GE)	%	55.0 (Africa)	Eq. 10.14–16; Annex 10A	Normal
`Ym`	Methane conversion factor	%	6.5 (cattle on forage)	Eq. 10.21; Table 10.12	Normal/PERT
`UE`	Urinary energy (fraction of GE)	—	0.04 (cattle)	Eq. 10.24 (footnote)	Constant
`ASH`	Ash content of manure (fraction of DMI)	—	0.08 (cattle; IPCC example 0.06 is pig-specific)	Eq. 10.24	Constant

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Table 1 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
MMS_fractions	Fraction of manure in each MMS (deterministic; constrained to sum to 1 per sub-category). Optional per-row lower_fraction / upper_fraction / distribution_fraction columns on the Manure_Management sheet enable per-iteration sampling within the user's bounds, renormalised to preserve the simplex; sampled fractions surface as fraction_<mms> in the sensitivity tornado.	—	country-specific (Worksheet 10.A-9)	Eq. 10.23	Deterministic (default) or per-MMS PERT/Normal/etc. with renormalisation
Bo	Maximum CH_4 producing capacity	m^3 CH_4/kg VS	0.13 (2019R "Other regions, low productivity" cattle); 2006 Africa = 0.10	Eq. 10.23; Table 10.16(a) (Updated)	Normal

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Table 1 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`MCF`	Methane conversion factor (MMS)	—	per-MMS & climate (e.g. 80% lagoon tropical)	Eq. 10.23; Table 10.17 (Updated)	Normal/PERT
`EF3`	Direct-N_2O EF (per MMS)	kg N_2O-N/kg N	per-MMS (e.g. 0.005 solid storage)	Eq. 10.25; Table 10.21	Lognormal/PERT
`Frac_GasMS`	Fraction N volatilised (MMS)	—	per-MMS, e.g. 0.45 solid storage (2019R)	Eq. 10.26; Table 10.22 (Updated)	PERT (preferred — IPCC ranges are asymmetric) / Normal
`EF4`	EF for atmospheric deposition	kg N_2O-N/(kg NH_3-N+NO_x-N)	0.010 aggregated 2019R (wet 0.014 / dry 0.005); 2006 = 0.010	Eq. 10.27; Vol.4 Ch.11 Table 11.3 (Updated)	PERT
`Frac_LeachMS`	Fraction N leached (MMS)	—	per-MMS, e.g. 0.02 solid storage	Eq. 10.28; Table 10.23 (2019R)	PERT

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Table 1 continued

Variable	Definition	Unit	IPCC default	IPCC reference (equation / table)	Typical dist.
`EF5`	EF for leaching/runoff	kg N_2O-N/kg N	0.011 (2019R); 2006 = 0.0075	Eq. 10.29; Vol.4 Ch.11 Table 11.3 (Updated)	PERT
`Frac_PRP`	Fraction of dung and urine deposited on pasture	—	country-specific	Vol.4 Ch.11 Eq. 11.5 ($MS_{(T,PRP)}$)	Beta / PERT (assignable via the `distribution` column)
`EF3_PRP`	Direct N_2O EF, pasture (cattle/poultry/pigs)	kg N_2O-N/kg N	0.004 aggregated 2019R (wet 0.006 / dry 0.002); 2006 = 0.02	Vol.4 Ch.11 Eq. 11.1; Table 11.1 (Updated)	PERT
`Frac_GasPRP`	Fraction N volatilised (PRP)	—	0.21 (2019R Frac-GASM); 2006 = 0.20	Vol.4 Ch.11 Eq. 11.9; Table 11.3 (Updated)	PERT (preferred — IPCC ranges are asymmetric) / Normal
`Frac_LeachPRP`	Fraction N leached (PRP)	—	0.24 (2019R wet); 2006 = 0.30; dry = 0	Vol.4 Ch.11 Eq. 11.10; Table 11.3 (Updated)	PERT

3 IPCC Tier 2 Emission Equations

This section presents the complete equation chain implemented in the tool, following IPCC (2006) Volume 4, Chapter 10. All equations are evaluated for each cattle sub-category independently; results are then aggregated to national totals by summing across sub-categories.

3.1 Net Energy for Maintenance

The daily net energy requirement for maintenance (NE_m, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.3:

$$NE_m = C_{fi,\text{adj}} \times BW^{0.75}$$

where BW is the mean live body weight (kg head^{-1}) and C_{fi} is a sex/condition coefficient (MJ kg$^{-0.75}$ day^{-1}). For open-lot cattle in colder climates the 2006 Guidelines (Vol.4 Ch.10, retained unchanged in the 2019 Refinement) prescribe a linear adjustment of C_{fi} — IPCC Eq.~10.2, “Coefficient for calculating net energy for maintenance’ ’ — based on the mean daily winter temperature:

$$C_{fi,\text{adj}} = C_{fi} + 0.0048 \times (20 - T_w) \quad \text{when } T_w < 20°\text{C}$$

otherwise $C_{fi,\text{adj}} = C_{fi}$. This adjustment increases the maintenance energy requirement by approximately 0.48% per degree below 20°C. The same linear formula is implemented in the IPCC Inventory Software (v2.95).

3.2 Net Energy for Activity

The net energy for physical activity (NE_a, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.4:

$$NE_a = C_a \times NE_m$$

where C_a is a dimensionless activity coefficient that accounts for the terrain and movement patterns of the animal (e.g. 0.00 for housed animals, 0.17 for animals on flat pasture, 0.36 for large areas or hilly terrain).

3.3 Net Energy for Growth

The net energy required for growth (NE_g, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.6:

$$NE_g = 22.02 \times \left(\frac{BW}{C \times MW}\right)^{0.75} \times WG^{1.097}$$

where WG is the mean daily weight gain (kg head^{-1} day^{-1}), MW is the mature body weight (kg), and C is a sex/physiological condition coefficient (0.8 for females, 1.0 for castrates, 1.2 for bulls). $NE_g = 0$ when $WG \leq 0$.

3.4 Net Energy for Lactation

The daily net energy for milk production (NE_l, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.8:

$$NE_l = \text{Milk} \times (1.47 + 0.40 \times \text{MilkFat})$$

where Milk is the daily milk yield (kg head^{-1} day^{-1}) and MilkFat is the fat content (%). For non-lactating sub-categories, set Milk = 0.

3.5 Net Energy for Work

The net energy for work (NE_w, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.11 (used when cattle perform physical work, e.g. traction or load-carrying):

$$NE_w = 0.10 \times NE_m \times \text{WorkHours}$$

where WorkHours is the mean number of hours of work per day. Set to 0 when animals do not perform work.

3.6 Net Energy for Pregnancy

The net energy allocated to fetal growth during pregnancy (NE_p, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.13:

$$NE_p = C_p \times NE_m$$

where C_p is a coefficient for the energy cost of pregnancy (0.10 for cattle). The parameter `pct_pregnant` can pro-rate C_p when not all females in a sub-category are pregnant in the same year ($0 \leq \text{pct\_pregnant} \leq 1$). Includes pregnant heifers that have not yet calved. The earlier name `pct_calving` is accepted as an alias on upload.

3.7 Energy Ratios: REM and REG

The ratio of net energy for maintenance to digestible energy consumed (REM) is given by IPCC Eq. 10.14:

$$REM = 1.123 - (4.092 \times 10^{-3} \times DE) + (1.126 \times 10^{-5} \times DE^2) - \frac{25.4}{DE}$$

The ratio of net energy for growth to digestible energy consumed (REG) is given by IPCC Eq. 10.15:

$$REG = 1.164 - (5.160 \times 10^{-3} \times DE) + (1.308 \times 10^{-5} \times DE^2) - \frac{37.4}{DE}$$

where DE is the digestible energy expressed as a percentage of gross energy. Both equations are undefined at $DE = 0$ (division by zero) — the tool validates $DE > 0$ before running any simulation.

3.8 Gross Energy Intake

The mean daily gross energy intake (GE, MJ head^{-1} day^{-1}) is given by IPCC Eq. 10.16:

$$GE = \frac{(NE_m + NE_a + NE_l + NE_w + NE_p)/REM + NE_g/REG}{DE/100}$$

3.9 Enteric Fermentation Emission Factor

The per-head per-day methane emission factor for enteric fermentation (EF_1, kg CH_4 head^{-1} day^{-1}) is given by IPCC Eq. 10.21:

$$EF_1 = \frac{GE \times (Y_m/100)}{55.65}$$

where Y_m is the methane conversion factor expressed as a percentage of gross energy converted to methane (the /100 converts Y_m to a fraction) and 55.65 MJ kg^{-1} is the energy content of methane. The annual per-sub-category enteric emission is:

$$E_{\text{enteric}} = EF_1 \times N \times 365/1000 \quad (\text{tonnes CH}_4\ \text{yr}^{-1})$$

3.10 Nitrogen Excretion

The daily nitrogen excretion rate (N_{ex}, kg N head^{-1} day^{-1}) follows IPCC Eq. 10.31:

$$N_{\text{ex}} = N_{\text{intake}} - N_{\text{retained}}$$

Nitrogen intake from feed is:

$$N_{\text{intake}} = \frac{GE}{18.45} \times \frac{\text{CP\%}}{6.25 \times 100}$$

where CP% is the crude-protein content of the diet (the template column is labelled `CP`; the value supplied is the same percentage value used here).

Nitrogen retained in body tissue and milk are calculated separately. For milk, the nitrogen content uses the **Jones factor for casein (6.38)** — distinct from the generic protein factor (6.25) used for feed and body tissue. The 6.38 factor and the 6.25 factor both appear inside IPCC 2006 Vol.4 Ch.10

Eq. 10.33 (N retention rates for cattle); the same equation is retained, with updated form, in the 2019 Refinement (also numbered 10.33):

$$N_{\text{milk}} = \frac{\text{Milk} \times \text{MilkProtein\%}}{6.38 \times 100}$$

For growing animals, IPCC Eq. 10.33 adds a body-tissue retention term:

$$N_{\text{growth}} = \frac{WG \times (268 - 7.03 \times NE_g/WG)}{1000 \times 6.25}$$

so that $N_{\text{retained}} = N_{\text{milk}} + N_{\text{growth}}$. For non-growing sub-categories ($WG \leq 0$) the growth term is set to zero.

The annual nitrogen excreted per sub-category ($N_{\text{ex,annual}}$, kg N yr^{-1}) is:

$$N_{\text{ex,annual}} = N_{\text{ex}} \times N \times 365$$

3.11 Volatile Solids and Manure CH_4

Total manure CH_4 emissions are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.22 (the top-level emissions equation, EF × population × 10^{-6}). The per-head per-year emission factor is given by Eq. 10.23:

$$EF_{\text{mm,CH}_4} = VS \times 365 \times B_o \times 0.67 \times \sum_s (MS_s \times MCF_s) \quad \text{(Eq. 10.23)}$$

The daily volatile-solid excretion rate VS (kg dry-matter head^{-1} day^{-1}) that feeds Eq. 10.23 is given by IPCC Eq. 10.24:

$$VS = [GE \times (1 - DE/100) + (UE \times GE)] \times \left(\frac{1 - \text{ASH}}{18.45}\right) \quad \text{(Eq. 10.24)}$$

where UE is the fraction of GE excreted as urinary energy (default 0.04 for cattle, IPCC Eq. 10.24 footnote) and ASH is the ash content of manure (default 0.08 for cattle).

where B_o is the maximum methane producing capacity (m^3 CH_4 kg^{-1} VS), 0.67 is the conversion factor from m^3 CH_4 to kg CH_4 (the density of methane at standard conditions, kg m^{-3}), MS_s is the fraction of manure managed in system s, and MCF_s is the methane conversion factor for system s (fraction of B_o). The 2019 Refinement Table 10.17 (Updated) provides MCF defaults across 11 manure management systems — eight inherited from the 2006 Guidelines plus four added in 2019 (anaerobic digester / biogas, aerobic treatment, solid storage – covered/compacted, and burned for fuel). Default B_o values for cattle are given in Table 10.16a. Per-MMS uncertainty in MCF and in the manure fractions MS_s is sampled independently — when the user supplies optional `lower_fraction` / `upper_fraction` / `distribution_fraction` columns on the Manure_Management sheet, each

Monte Carlo iteration's MS_s row is renormalised after sampling so the simplex constraint $\sum_s MS_s = 1$ is preserved exactly.

3.12 Manure Management N_2O (Direct)

Direct N_2O emissions from manure management are given by IPCC 2006 Vol.4 Ch.10 Eq. 10.25 (the indirect-pathway losses are calculated separately in Eq. 10.26–10.29; see next sub-section):

$$E_{\text{direct,mm}} = N_{\text{ex,annual}} \times EF_3 \times \frac{44}{28} \quad (\text{kg } N_2O \text{ yr}^{-1})$$

where EF_3 is the emission factor for direct N_2O (kg N_2O-N kg^{-1} N) and 44/28 converts N_2O-N to N_2O. For multiple MMS systems, EF_3 is computed as the weighted average of per-MMS factors:

$$EF_3 = \sum_s MS_s \times EF_{3,s}$$

3.13 Manure Management N_2O (Indirect)

Indirect N_2O from managed manure arises from two pathways. Each pathway is split between an N-loss equation (in Vol.4 Ch.10) and an N_2O equation (in Vol.4 Ch.10, drawing EFs from Vol.4 Ch.11 Table 11.3):

Atmospheric deposition of volatilised N — Frac_GasMS from Eq. 10.26 multiplied by EF_4 in Eq. 10.27:

$$E_{\text{vol,mm}} = N_{\text{ex,annual}} \times \text{Frac\_GasMS} \times EF_4 \times \frac{44}{28}$$

Leaching and runoff — Frac_LeachMS from Eq. 10.28 multiplied by EF_5 in Eq. 10.29:

$$E_{\text{leach,mm}} = N_{\text{ex,annual}} \times \text{Frac\_LeachMS} \times EF_5 \times \frac{44}{28}$$

where EF_4 (kg N_2O-N kg^{-1} NH_3-N + NO_x-N volatilised) and EF_5 (kg N_2O-N kg^{-1} N leached/runoff) are from IPCC Vol.4 Ch.11 Table 11.3. In the 2006 Guidelines $EF_4 = 0.010$ (range 0.002–0.05) and $EF_5 = 0.0075$ (range 0.0005–0.025). The 2019 Refinement keeps the aggregated $EF_4 = 0.010$ (range 0.002–0.018) and additionally provides climate-disaggregated values of 0.014 in wet climates (range 0.011–0.017) and 0.005 in dry climates (range 0.000–0.011); EF_5 is updated to 0.011 (range 0.000–0.020) with no climate disaggregation. Frac_GasMS and Frac_LeachMS are MMS- and country- or climate-specific fractions (defaults in Vol.4 Ch.10 Tables 10.22 and 10.23 of the 2019 Refinement).

3.14 Pasture/Range/Paddock N_2O (Direct)

The nitrogen deposited on pasture by grazing animals (F_{PRP}) is the per-sub-category form of IPCC Vol.4 Ch.11 Eq.~11.5 (N in urine and dung deposited by grazing animals on PRP):

$$F_{\text{PRP}} = N_{\text{ex,annual}} \times \text{Frac\_PRP}$$

where Frac_PRP is the fraction of total annual N excretion deposited on pasture (denoted $MS_{(T,\text{PRP})}$ in IPCC notation). Direct N_2O from pasture/range/paddock falls under managed soils (IPCC Volume 4, Chapter 11), not Chapter 10. It is given by IPCC Eq.~11.1 (with F_{PRP} as the input N source):

$$E_{\text{direct,PRP}} = F_{\text{PRP}} \times EF_{3,\text{PRP}} \times \frac{44}{28}$$

where $EF_{3,\text{PRP}}$ is the emission factor for direct N_2O from PRP given in IPCC Vol.4 Chapter 11, Table 11.1. The 2006 value is 0.02 kg N_2O-N kg^{-1} N (cattle, poultry, pigs combined). The 2019 Refinement updates this to $EF_{3PRP,CPP}$ for cattle, poultry and pigs with an aggregated default of 0.004 kg N_2O-N kg^{-1} N (range 0.000–0.014). When the climate is known, the disaggregated values are 0.006 in wet climates (range 0.000–0.027) and 0.002 in dry climates (range 0.000–0.007).

3.15 Pasture N_2O (Indirect)

Indirect N_2O from pasture deposition follows IPCC Vol.4 Ch.11 Eq. 11.9 (volatilisation pathway) and Eq. 11.10 (leaching/runoff pathway), using the same pathway structure as managed manure:

$$E_{\text{indirect,PRP}} = F_{\text{PRP}} \times (\text{Frac\_GasPRP} \times EF_4 + \text{Frac\_LeachPRP} \times EF_5) \times \frac{44}{28}$$

On the soil side of Eq.~11.9/11.10 the same role played by Frac_GasMS and Frac_LeachMS for managed manure is played by FracGASM and FracLEACH-(H) in IPCC Vol.4 Ch.11 Table 11.3 — Frac_GasPRP corresponds to FracGASM (the fraction of applied/deposited N volatilised as NH_3 and NO_x) and Frac_LeachPRP corresponds to FracLEACH-(H) (the fraction leached/runoff in regions with annual precipitation exceeding evapotranspiration).

Per IPCC Vol.4 Ch.11, where evapotranspiration exceeds precipitation, leaching may be assumed not to occur — in the tool, `Frac_LeachPRP` may be set equal to zero in those climate cases.

3.16 GWP-Based CO_2eq Conversion

All emission pathways are converted to CO_2 equivalents using Global Warming Potentials (GWP). The tool supports three IPCC assessment report baselines:

Basis	**CH_4 GWP**	**N_2O GWP**	**Reference**
AR4 (100-yr)	25	298	IPCC AR4 WG1, Ch.2 Table 2.14
AR5 (100-yr)	28	265	IPCC AR5 WG1, Ch.8 Table 8.7
AR6 (100-yr, non-fossil CH_4)	27.0	273	IPCC AR6 WG1, Ch.7 Table 7.15 (and Table 7.SM

Total CO_2eq = (total CH_4 tonnes × GWP_{CH_4}) + (total N_2O tonnes × $\text{GWP}_{\text{N}_2\text{O}}$).

4 Monte Carlo Uncertainty Analysis

4.1 IPCC Approach 2 Rationale

The IPCC (2006) Volume 1, Chapter 3 defines two approaches to quantifying inventory uncertainty:

- **Approach 1 (error propagation):** Combines individual parameter uncertainties using simple algebraic rules. Suitable when the emission equation is a product or sum of independent terms. Underestimates uncertainty when equations are non-linear or inputs are correlated.
- **Approach 2 (Monte Carlo simulation):** Propagates uncertainty through the full equation chain by repeatedly sampling all parameters from their distributions. Exact for any equation structure. Recommended for Tier 2 livestock inventories because the GE equation chain is non-linear in BW, WG, and $DE_{\%}$.

This tool implements Approach 2. The simulation samples all input parameters simultaneously for each iteration, evaluates the full equation chain from NE_m through to total CO_2eq, and records the output. The resulting distribution of outputs characterises the uncertainty.

4.2 Probability Distributions

Each parameter is assigned a probability distribution reflecting the state of knowledge about its true value. The tool supports:

Distribution	Parameterisation	Typical use
Normal	mean μ, SD σ	Body weight, milk yield, population counts with symmetric survey error
Positive-Normal	Normal truncated at zero	Non-negative symmetric parameters where negative draws are implausible
Lognormal	log-mean, log-SD (fitted from mean and CV%)	Strictly positive, right-skewed; milk yield in heterogeneous herds; some EFs
Beta	rescaled Beta(α, β) matching the supplied mean and bounds	Fractions in $[0, 1]$ with mode strictly in the interior; pct_pregnant, Cp
Truncated-Normal [0,1]	Normal clipped to $[0, 1]$	Fractions in $[0, 1]$ when a symmetric error around the mean is appropriate
PERT	minimum, most-likely, maximum	Emission factors with expert-elicited bounds; EF3_PRP, EF4, EF5, Bo, MCF
Triangular	minimum, most-likely, maximum	Similar to PERT but with sharper tails; Ca, C_growth
Uniform	lower, upper	Parameters with only the plausible range known; MMS fractions
Constant	fixed value	Parameters treated as known exactly (uncertainty_pct = 0)

When the user supplies `uncertainty_pct` (a symmetric 95% half-width as a percentage of the mean), the tool derives the distribution parameters as:

$$\sigma = \frac{\text{uncertainty\_pct} \times \mu}{100 \times 1.96}$$

for a normal distribution, or equivalent transformations for lognormal and PERT.

4.3 Correlated Sampling

When parameters are correlated (e.g. body weight and mature weight are positively correlated; diet quality and methane conversion factor may be negatively correlated), ignoring correlation underestimates or overestimates combined uncertainty. The tool applies the same correlated-sampling procedure to every path that supplies a correlation matrix — the structural-defaults preset, a manually uploaded matrix, and the matrix estimated automatically from a time-series upload all flow through the same algorithm.

Procedure (per IPCC Volume 1 Chapter 3 §3.2.3.2):

1. For each parameter j, draw n_{iter} independent samples from the parameter's own marginal distribution (normal, PERT, lognormal, beta, triangular, uniform, or constant, exactly as

specified in the template).

2. Reorder the columns of the resulting $n_{\text{iter}} \times p$ sample matrix so that the realised Spearman rank correlation between columns matches the target matrix $\boldsymbol{\Sigma}$ (implemented via `mc2d::cornode`, which performs the restricted-pairing operation IPCC §3.2.3.2 cites under "restricted pairing methods that are included in many software packages' ').

This procedure is distribution-free: the marginal shapes are preserved exactly, regardless of whether the user has mixed normal / lognormal / PERT / beta parameters in the same correlation block. The target $\boldsymbol{\Sigma}$ is interpreted as a Spearman rank correlation matrix throughout — the same convention the time-series matrix is computed under (Spearman rank correlation of the user's annual values, after detrending; see Section "Correlation Structure' ').

If $\boldsymbol{\Sigma}$ is not positive definite (e.g. a manually entered matrix that turns out to be inconsistent), the tool projects it onto the nearest positive-definite correlation matrix via `Matrix::nearPD(corr = TRUE)` and emits a warning whenever any individual entry is shifted by more than 0.01. The tool does *not* silently fall back to uncorrelated sampling.

The tool supports both within-block and across-block correlations. The default scope is independence; a single unified correlation matrix spanning AD + coefficients is used whenever cross-block pairs are specified.

4.4 Simulation Procedure

For each Monte Carlo iteration $i = 1, \ldots, n_{\text{iter}}$:

1. Sample all parameters from their distributions (with correlation structure if supplied).
2. For each cattle sub-category k:
   a. Compute $NE_m^{(k)}$, $NE_a^{(k)}$, $NE_g^{(k)}$, $NE_l^{(k)}$, $NE_w^{(k)}$, $NE_p^{(k)}$.
   b. Compute $REM^{(k)}$ and $REG^{(k)}$ from $DE_{\%}^{(k)}$.
   c. Compute $GE^{(k)}$, $EF_1^{(k)}$, $N_{\text{ex}}^{(k)}$, $VS^{(k)}$.
   d. Compute emission totals for all six pathways.
3. Sum across all sub-categories to obtain national pathway totals and total CO_2eq.
4. Record all outputs for iteration i.

After all iterations, compute summary statistics from the distribution of recorded outputs.

4.5 Recommended Iteration Count and Convergence

Monte Carlo estimates converge as $n_{\text{iter}} \to \infty$. For practical use:

- **1,000 iterations** — suitable only for a quick test run to verify the model runs without errors. Do not report results from 1,000 iterations.
- **10,000 iterations** — minimum recommended for any result intended for use. Convergence is generally adequate for inventories with up to 5 sub-categories and 30 parameters.
- **25,000–30,000 iterations** — recommended for final reporting, especially when correlations

are enabled or the inventory has many sub-categories or asymmetric distributions.

- **50,000 iterations** — high-precision run; use when the 95% CI bounds need to be stable to two significant figures.

Convergence check: Re-run with a different random seed (e.g. 123 and 456). If the 95% MoE changes by more than 1–2 percentage points, increase the number of iterations.

5 Uncertainty Metrics

The following metrics are reported for each emission pathway and for the total CO_2eq:

5.1 95% Confidence Interval

The 95% CI is defined as the 2.5th and 97.5th percentiles of the Monte Carlo output distribution:

$$\mathrm{CI}_{95} = [Q_{0.025},\ Q_{0.975}]$$

This is a simulation-based (quantile) interval — it does not assume a normal distribution and is therefore valid for asymmetric emission distributions.

5.2 Margin of Error (the IPCC reporting metric)

The margin of error (MoE%) is the half-width of the 95% CI expressed as a percentage of the mean:

$$\mathrm{MoE\%} = \frac{Q_{0.975} - Q_{0.025}}{2 \times \bar{x}} \times 100$$

MoE% is the metric IPCC Vol.1 Ch.3 Table 3.3 requires ("% uncertainty" in that table is defined as the half-width of the 95% CI divided by the mean). It is the headline number reported in the user-facing value boxes, in the IPCC Report tab, and in the Excel / Word downloads. Because MoE is derived from the empirical 2.5% / 97.5% quantiles of the Monte Carlo output, it correctly captures distribution skewness — for a perfectly normal distribution it reduces to $\approx 1.96 \times \mathrm{CV\%}$, but for skewed distributions (lognormal EFs, PERT-bounded fractions, manure-N_2O sums) MoE and $1.96 \cdot \mathrm{CV}$ differ.

5.3 Coefficient of Variation (kept for reference, not reported as headline)

The coefficient of variation (CV%) expresses uncertainty relative to the mean:

$$\mathrm{CV\%} = \frac{\sigma}{\bar{x}} \times 100$$

where σ is the standard deviation and $\bar{x}$ is the mean. CV% is **still computed internally** for every emission source (kept in the underlying `calc_all_uncertainty()` data frame for advanced reproducibility) but it is no longer surfaced in any user-facing display — IPCC Annex 7 expects MoE%, not CV%. It is undefined (reported as NA) when the mean emission is zero (e.g. pasture N_2O for fully housed systems).

5.4 IPCC Table 3.3 Format

IPCC (2006) Volume 1, Table 3.3 requires reporting uncertainty for each emission category as follows:

Source / Sink Category	Emission (t CH_4 or t N_2O)	AD Uncertainty (95% MoE %)	EF Uncertainty (95% MoE %)	Combined Uncertainty (95% MoE %)
Enteric fermentation CH_4	...	MoE% from AD-only run	MoE% from EF-only run	MoE% from combined run
...				

The tool generates this table automatically after each simulation run. The Excel download labels each uncertainty column explicitly as "(95% MoE %)" so there is no ambiguity with CV.

6 Uncertainty Decomposition

To separate the contributions of activity data (AD) uncertainty and emission factor (EF) uncertainty to the combined uncertainty, the tool performs three simulation runs:

1. **Combined run** — all parameters varied according to their distributions. This is the primary result.
2. **AD-only run** — emission factors fixed at their central values ($\sigma = 0$); only AD parameters vary. 95% MoE from this run = AD uncertainty contribution.
3. **EF-only run** — AD parameters fixed at their central values; only emission factors vary. 95% MoE from this run = EF uncertainty contribution.

This approach correctly handles multi-group inventories by running all three simulations through the full multi-sub-category equation chain and aggregating to national totals before computing metrics.

The decomposition reveals whether uncertainty reduction efforts should focus on improving animal population surveys (AD) or on refining emission factor estimates (EF).

Implementation note. The AD-only run pins ALL coefficient-side variance, not only the Parameters-sheet rows. The per-MMS sample matrices used for MCF, EF3, Frac_GasMS, Frac_LeachMS, and (when supplied) the MMS-allocation MS_s fractions are nulled out for the AD-only sub-run so iteration-to-iteration variation in those quantities cannot leak into the AD-only result. As a result, AD-only CV equals CV(N) for every emission source in a single-sub-category inventory and varies across sources only through the Herfindahl-like concentration of each source over multiple sub-categories.

7 Sensitivity Analysis

Sensitivity analysis identifies which input parameters have the greatest influence on the total emission estimate. The tool implements two complementary methods.

7.1 Standardised Regression Coefficients (SRC)

SRC is based on fitting a multiple linear regression of the standardised output on all standardised inputs:

$$Y^* = \beta_0 + \sum_j \beta_j X_j^* + \varepsilon$$

where $Y^* = (Y - \bar{Y})/\sigma_Y$ and $X_j^* = (X_j - \bar{X}_j)/\sigma_{X_j}$. The SRC for parameter j is the coefficient β_j from this regression.

Properties: - A positive SRC indicates that higher values of the parameter produce higher emissions. - The magnitude $|\beta_j|$ reflects the relative importance of parameter j. - SRC values range from approximately -1 to $+1$ when a single parameter dominates. - The R^2 of the regression indicates what fraction of the total variance in the output is explained by the linear model. High R^2 (> 0.9) indicates SRC is a reliable measure; low R^2 suggests non-linearity. - SRC requires only a single regression fit, making it computationally efficient for any number of parameters.

7.2 Partial Rank Correlation Coefficients (PRCC)

PRCC is a rank-based method that is robust to non-linearity and non-normal distributions:

1. Replace all input columns and the output column with their ranks.
2. For each parameter j:
   a. Regress rank(X_j) on all other ranked inputs; compute residuals e_{X_j}.
   b. Regress rank(Y) on all other ranked inputs; compute residuals e_Y.
   c. $\text{PRCC}_j = \text{cor}(e_{X_j},\ e_Y)$.

PRCC measures the unique rank-correlation between each input and the output after removing the shared linear effect of all other inputs. It is more appropriate than SRC when the relationship between inputs and output is monotone but non-linear, or when input distributions are strongly skewed.

Computational note: PRCC requires one regression fit per parameter. For large inventories (more than 40 combined parameters across all sub-categories), PRCC computation can take several minutes. In this case, the tool automatically falls back to SRC for the aggregate sensitivity result; PRCC remains available for individual sub-category analyses.

7.3 Tornado Chart Interpretation

Each parameter label on the tornado chart is formatted as <parameter> (<sub_category>) — for example `Ym (DINT_cow)`, `BW (DINT_heif)`, `MCF_solid_storage (DINT_cow)`. The sub-category in parentheses identifies which animal group each influential parameter belongs to so that reduction efforts can be targeted at the specific sub-category that drives uncertainty rather than at the parameter in aggregate. The same labels appear in the rank-correlation table and the downloadable Word / Excel sensitivity sheets.

The tornado chart displays the top parameters ranked by |SRC| or |PRCC|:

- **Green bars** (positive coefficient) — increasing this parameter increases total emissions; reduction in uncertainty would directly reduce total emission uncertainty.
- **Red bars** (negative coefficient) — increasing this parameter decreases total emissions.
- **Dark (slate) bars** — IPCC-defined coefficients that cannot be improved by the national inventory compiler (e.g. GWP values, Cfi, Cp). Uncertainty in these parameters is irreducible without IPCC guidance changes.
- **Bright (green/red) bars** — user-reducible parameters where better national data could meaningfully reduce inventory uncertainty.

8 Correlation Structure

8.1 Activity Data Correlations

Correlations between activity data parameters (e.g. body weight and mature weight, milk yield and milk fat content) are entered in the Tab~4 *Activity-Data Correlations* card. Four modes are available; each is described in detail below.

- **No correlations** (independence) — the default. All parameters are drawn independently. This is the conservative IPCC Approach 2 starting point and is appropriate when the inventory compiler has no defensible information about parameter co-movement.
- **From template (auto, time-series)** — the matrix is computed automatically from the `Parameter_TimeSeries` sheet of the upload using a **Spearman rank correlation** after detrending (first differences by default; linear and raw alternatives available). Spearman is preferred over Pearson because (i) it is robust to skewed marginals (Ym, Bo, milk yield), (ii) it matches the rank-correlation target the sampler reproduces (per IPCC Vol.1 Ch.3 §3.2.3.2), and (iii) it is the preferred option when the user wants the *rank* dependence preserved rather than a linear product-moment relationship that can be distorted by outliers. Requires ≥ 5 annual rows and ≥ 2 non-constant numeric columns to produce a usable matrix; otherwise the option is greyed out (see “UI gating” below).
- **Structural-defaults preset** — a small set of expert-elicited pairs drawn from the IPCC equation structure and the published livestock literature (see Section “Structural-Defaults Preset” below). Independent of the upload’s time-series, so this is the option that works for single-year inventories without historical data.
- **Manual entry** — the user uploads a square symmetric CSV. The diagonal must be 1; entries are interpreted as Spearman rank correlations on $[-1, +1]$; the matrix is projected onto the nearest positive-definite correlation matrix if needed, with a warning if any entry shifts by more than 0.01.

UI gating. Modes whose prerequisites are missing are greyed out automatically and cannot be selected: `From template'' is disabled when the upload has no usable \texttt{Parameter\_TimeSeries} sheet,`Manual entry’ ’ is disabled until the user uploads a CSV matrix, and `Structural defaults'' is disabled until a template (built-in example or custom upload) is loaded. The Run button additionally blocks the simulation with an explicit error if a no-op mode somehow slips through. These gates prevent the silent fall-back to`no correlations’ ’ that would otherwise occur when a mode is selected without the data it needs.

Expected magnitude of effect on the headline. Picking the structural-defaults preset on a typical single-year inventory widens the 95% uncertainty interval by approximately 3–10%, depending mainly on how wide the user’s Ym range is (the cross-block DE $\leftrightarrow$ Ym $= -0.50$ pair is the dominant mover). The central estimate is essentially unchanged because correlations preserve marginal means.

The “From template (auto)’ ’ mode produces effects in the same general range when the time-series is rich enough to deliver multiple correlated pairs; a sparse or near-static time-series will produce a much smaller effect (the matrix collapses to near-identity after detrending).

8.2 Emission Factor Correlations

Emission factors that share systematic bias because they come from the same measurement literature (e.g. all rumen-fermentation coefficients from one regional database, or all NH_3 volatilisation parameters from one experimental programme) can be assigned correlations in the Tab~4 *Coefficient Correlations* card. Two modes are available:

- **No EF correlations** (default) — coefficients are sampled independently. The standard IPCC Approach 2 assumption and the right default when each emission factor comes from its own study.
- **Block-structured EF correlation** — coefficients are grouped into three blocks (energy-equation, manure-CH_4, manure-N), each with its own within-block ρ slider in $[0, 0.5]$. Cross-block correlation is always zero because the three measurement literatures are independent. All three sliders default to 0, so this mode does nothing until the user actively moves at least one slider; an inline warning surfaces if the user selects “block-structured” but leaves all sliders at zero.

The same correlated-sampling procedure used for activity data (Section “Correlated Sampling’ ’ above) is applied to the coefficient block — no special-case sampler. Setting a within-block slider to $\rho = 0.3$ typically widens the 95% uncertainty interval by a further few percent on top of any activity-data correlation effect; setting it to $\rho = 0.5$ (the slider maximum) approximately doubles that.

8.3 Structural-Defaults Preset (Expert-Elicited)

The tool provides a **structural-defaults preset** correlation matrix as a middle ground between full independence and user-specified correlations. This preset assigns non-zero correlations to a small set of structurally linked pairs drawn from the IPCC equation structure and the published livestock literature. *These values are expert-elicited; the IPCC publishes no numerical correlation coefficients for any livestock parameter pair.* All other pairs are set to zero. Hover any non-zero cell in the heatmap for the per-pair source citation. This is appropriate when no empirical correlation data are available but the inventory compiler wants to avoid the unrealistic assumption of full independence.

The current preset contains seven pairs:

Pair	ρ	**Source / rationale**
BW ↔ MW	+0.50	Growth-curve relation (Brody 1945) reused in IPCC Eq 10.3. The high same-survey value 0.85 only holds when BW and MW are measured together; in most national inventories MW is taken from a breed reference table while BW comes from the livestock census, so the cross-source correlation is closer to 0.5.
BW ↔ WG	+0.40	IPCC Eq 10.6: $NE_g \propto BW^{0.75} \cdot WG^{1.097}$; joint regression-estimation correlation in NRC (2001) energy-balance studies, published range 0.30–0.50. Sign is system-dependent (heavy growing animals gain slower ⇒ negative; well-fed adults are both heavier and gain more ⇒ positive). The +0.40 default reflects the well-fed-adult case typical of dairy/beef finishing systems.
Milk ↔ Fat	−0.30	Milk dilution effect, dairy genetics literature (Wilmink 1987; VandeHaar 1998). Within-herd literature gives −0.20 to −0.40; −0.30 is the midpoint.
Milk ↔ BW	+0.30	High-producing dairy breeds are physically larger: Holstein ∼650 kg, ∼30 kg milk/day vs Jersey ∼450 kg, ∼20 kg milk/day. NRC Dairy (2001); Stallings & Knowlton (2013). Captures a real cross-group biological linkage.
Milk ↔ DE	+0.30	Higher feed digestibility ⇒ more nutrient availability ⇒ higher milk yield, especially in concentrate-fed dairy systems. NRC Dairy (2001) Ch.2.
DE ↔ CP	+0.50	Forage-quality co-variation across feed types (NRC 2001; INRA; Feedipedia). High-quality forages have both high DE and high CP; cross-feedstuff correlations range 0.4–0.7.
DE ↔ Ym	−0.50	The IPCC 2019 Refinement treats Ym as a decreasing function of diet digestibility (the updated Ym guidance in Table 10.12 and the Ym-prediction approach in Annex 10B.2 give lower Ym for higher-quality, more-digestible diets). Meta-analyses (Niu et al. 2018 *Global Change Biology*; Hristov et al. 2013) show a cross-diet negative correlation of this order. The single cross-block (activity-data ↔ coefficient) pair in the preset and the one most likely to move headline emissions.

Derivation notes. A previously included N ↔ BW = 0.30 pair was removed (no defensible crosscountry source); DE ↔ Ym was set to −0.50 (cross-diet meta-analyses cited in Section "Activity Data Correlations''); Cfi ↔ Ca was set to 0.30 rather than the previously assumed 0.60, recognising that Cfi and Ca are independent inputs in IPCC Eq 10.3 / 10.4 rather than jointly estimated; BW ↔ MW was set to 0.50 rather than 0.85 to reflect the typical mixed-source data provenance (BW from the national census, MW from a breed reference); the two cross-group dairy pairs (Milk ↔ BW and Milk ↔ DE) were added to capture biological linkages between herd-population and feed-quality parameters; and a Milk ↔ pct_pregnant pair was not included because the sign is ambiguous between per-cow genetic and per-herd management interpretations. The resulting matrix is positive-definite without nearest-PD repair.

8.4 Cross-Block Correlations

The preset and the manual / time-series matrices support cross-block correlations (i.e. correlations between an activity-data parameter and a coefficient — for example BW ↔ Cfi). This is implemented via a single unified correlation matrix that spans the union of all parameter names; the sampler draws from one multivariate distribution covering both blocks rather than two independent block-level distributions. Independence within and across blocks is the default.

8.5 Time-Series Correlation Matrix

When the user uploads a `Parameter_TimeSeries` sheet, the correlation matrix is computed automatically as a **Spearman** rank correlation of the annual values, after **detrending** (first differences by default, with a UI option to switch to linear detrend or raw series). Detrending separates shared long-run growth — which is not the year-to-year joint uncertainty Approach 2 is meant to capture — from genuine parameter co-movement. The resulting matrix is then projected onto the nearest positive-definite correlation matrix via `Matrix::nearPD(corr = TRUE)` so it can be supplied directly to the sampler described in the "Correlated Sampling'' section above.

8.6 Year-to-Year Coefficient Correlation (Trend Mode)

When the tool is run in **Trend mode** (multi-year inventory), an additional decision arises: within a single Monte Carlo iteration, how should the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, . . .) move from one year to the next? IPCC Volume 1 Chapter 3 Section 3.2.3.2 (Trend Analysis) and 2019 Refinement §3.2.2.4 discuss this explicitly: emission factor uncertainties are typically estimated once for an inventory series and reused across years (correlated), while activity data is normally re-estimated each year and treated as independent across years.

The tool exposes three options on the Simulate tab when Trend mode is selected. Activity data (N, BW, WG, Milk, . . .) is re-sampled fresh each year regardless of this setting:

Option	What happens inside one Monte Carlo iteration
Fully correlated (default)	The coefficient draw for year t is the *same* as for year 0. If Ym is drawn at 6.2% for year 0, it is 6.2% for every other year in that iteration. The trend then reflects activity-data changes only — coefficient uncertainty cancels when subtracting two years. Matches IPCC 2019 §3.2.2.4 default behaviour and the Ogle et al. (2003) "identical random seed" construction cited in V1 Ch3 Box 3.2.
Partial (AR(1), $\rho = 0.7$)	Coefficients drift slowly between years with a rank correlation of $\rho^{\lvert i-j \rvert}$ between year i and year j ($\rho = 0.7$). Independent samples are drawn for each year from the coefficient's marginal and then reordered across years so the realised Spearman rank correlation matches the AR(1) target — the same restricted-pairing procedure used within a year (per IPCC Vol.1 Ch.3 §3.2.3.2), applied across years. Neighbouring years share most of the same observational basis but the coefficient can move over a decade. Appropriate when the EF is re-estimated periodically (e.g. every 3–5 years).
Independent	Each year gets a fresh, independent coefficient draw. Maximises the EF contribution to trend uncertainty. Realistic only when every emission factor is genuinely re-measured with new field data every year, which is uncommon for national inventories.

The choice is implemented in `R/trend_tab.R`. The `full'' branch pre-samples a single $n_{\text{iter}} \times n_{\text{coef}}$ matrix and reuses it for every year. The`partial'' branch builds an $n_{\text{iter}} \times n_{\text{years}}$ rank-correlated draw for each coefficient against an AR(1) target rank correlation matrix, then runs each year's MC with the corresponding column slice. The "independent'' branch re-samples coefficients inside every per-year call.

Direction of effect. For an emission factor whose uncertainty is largely systematic, full correlation produces the *smallest* trend uncertainty (because the EF cancels) and independent sampling the *largest*. IPCC V1 Ch3 p. 25 explicitly notes that `positive correlations in uncertainties when comparing two years as part of trend analysis will decrease uncertainty in the trend''. Choosing`independent'' when the EFs are in fact reused will over-state trend uncertainty; choosing `full'' when the EFs are in fact re-measured each year will under-state it. The IPCC default (`full'') is the conservative choice for the typical case where the inventory compiler is using IPCC defaults or a single national estimation programme.

9 Validation and Quality Assurance

The tool includes a QA/QC module (`R/utils_qaqc.R`) that runs a series of checks against the loaded parameter data before simulation. Each check returns one of five statuses:

Status	Trigger	What the user sees
Pass	All structural checks pass. For `BW`, the mean is also within ±50% of the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3).	Green tick; no action needed.
Warn	For `BW`: value deviates 50–200% from the continental IPCC default. For any parameter: implausible bounds ($BW > 1500$ kg, $Y_m > 12\%$); a fractional parameter is using an unbounded distribution; or another non-fatal structural condition fired.	Amber row; the message either suggests a fix or asks the user to document the country-specific deviation.
Fail	For `BW`: value deviates more than 200% from the continental IPCC default. For any parameter: $DE_{\%} \leq 0$ or ≥ 100; or *lower* > *upper*; or an MMS fraction-sum is more than 1% off 100%; or a Beta is configured with mean outside $(0, 1)$.	Red row; simulation will still run but the result will be flagged in the report.
Missing	A core parameter was not supplied in the upload; auto-filled with the IPCC default. Applies to every parameter, not just `BW`.	Grey row; the message names the IPCC table or equation used and, for context-dependent parameters (EF3_PRP, EF4, Frac_LeachPRP, Bo, MCF, Ym, Cfi, Ca), appends an explicit hint of which alternative IPCC table row should be reviewed for the inventory's climate / production system / animal class.
Info	Reserved for future advisory messages.	Blue row.

The Pass / Warn / Fail thresholds for benchmark deviation apply **only to body weight (`BW`)**, the single parameter for which the tool holds a defensible continental IPCC default lookup: Vol.4 Ch.10

Annex Table 10A.1 (dairy cows), Annex Table 10A.2 (other cattle), or Annex Table 10A.3 (buffalo). Earlier versions of the tool also flagged deviations on Milk, MW, DE, Ym, and Bo against heuristic mid-points derived from the same Annex 10A illustrative tables; these were removed (Andreas Wilkes review, May 2026) because those mid-points were not directly citable to a published IPCC default and a reviewer auditing the QA tab could not locate them in the guidelines. The `ipcc_default` column of `PARAM_CATALOGUE` is still used to auto-fill missing values and is sourced from Vol.4 Ch.10 (Tables 10.4, 10.5, 10.7, 10.11, 10.12, 10.16(a), 10.17, 10.21, 10.22, 10.23) and Vol.4 Ch.11 (Tables 11.1, 11.3) — it is just no longer used as a deviation benchmark for parameters other than BW. Reinstating per-parameter benchmarks properly would require a multi-dimensional table indexed by continent, IPCC version (2006 vs 2019R), production system, and animal sub-category; that is deferred to a future release. Auto-filled parameters are listed in a banner on Tab~1 immediately after data upload; they should be replaced with country-specific values wherever possible, because both the central estimate and the propagated uncertainty depend on them.

10 Verification of Calculation Accuracy

The previous section describes the checks the tool applies to the user's *inputs*. This section describes how the tool's *calculation engine* — the implementation of the IPCC equation chain and the Monte Carlo sampler — has itself been verified to produce correct numbers. Input validation and calculation verification are complementary: the former guards against bad data going in, the latter guarantees the equations are implemented correctly.

10.1 Equation-chain verification against manual calculation

The core of the verification is a fixed, fully worked test case. For a single representative animal with a complete set of fixed parameters, every intermediate quantity in the IPCC Tier 2 chain was computed independently by hand and in a spreadsheet, directly from the published equations: the net-energy terms (NE_m, NE_a, NE_g, NE_l, NE_w, NE_p), the energy ratios (REM, REG), gross energy intake (GE), the enteric emission factor (EF_1), volatile-solid excretion (VS), nitrogen excretion (N_{ex}), and all four manure- and pasture-N_2O pathways, through to the inventory totals (total CH_4, total N_2O, and total CO_2eq under each GWP basis).

The tool's computed value for each of these quantities must match the independently calculated reference to a relative tolerance of 1×10^{-4}. These comparisons are encoded as automated assertions that run on every build, so any future change to the calculation code that altered a result — even slightly — would be caught immediately. This is the primary guarantee that the equations in Section "IPCC Tier 2 Emission Equations" are implemented exactly as written.

10.2 Cross-comparison with an independent Monte Carlo package

Beyond the single worked case, the tool has been exercised on real Tier 2 national cattle inventories and its results compared against two independent references: (i) manual re-calculation of the per-head emission factors and inventory totals, and (ii) an independent commercial Monte Carlo package configured with the same parameters, distributions and iteration count.

On these real inventories the tool reproduced the deterministic emission totals from the manual calculation, and produced a 95% uncertainty interval consistent with the independent Monte Carlo package to within the sampling variability expected between two stochastic runs. %% OPTIONAL: if you wish to quote a specific figure, insert it here, e.g. %% "(agreement within X% on the 95% margin of error for the manure-N2O pathway)". A regression guard retained in the automated test suite keeps the manure-N_2O pathway within the range established by this cross-comparison, so a future code change cannot silently introduce a large (e.g. order-of-magnitude) error in that pathway.

10.3 Automated regression test suite

The two verification activities above are part of a broader automated test suite (91 checks at the time of writing) that is run before each release and deployment. The checks are organised into seven groups:

- **A — IPCC equation chain:** every intermediate and total against the hand-calculated reference case (Section "Equation-chain verification" above).
- **B — Monte Carlo sampler:** empirical means of the sampled marginals match their analytical means; realised Spearman rank correlations match the requested targets; the AR(1) year-to-year correlation structure behaves as specified.
- **C — Application routes:** emission-source filtering (all sources, enteric only, manure CH_4 only, N_2O only), the three GWP bases (AR4 / AR5 / AR6), the correlation modes, and the AD / EF / combined uncertainty decomposition.
- **D — Trend (multi-year) pipeline:** the full / partial / independent year-to-year coefficient-correlation modes produce valid multi-year results.
- **E — Multi-sub-category aggregation:** per-sub-category emissions sum correctly to national totals and scale as expected with population.
- **F — Edge cases and validators:** missing-value handling, manure-management fraction bounds, lower/upper-bound consistency, zero-population handling, and the multi-system N_2O calculations.
- **G — Outputs:** the Excel and Word/CSV downloads and the correlation-matrix round-trip produce well-formed, internally consistent files.

The full multi-sub-category inventory chain must pass all checks before the tool is published.

10.4 Verification of IPCC default values and uncertainty ranges

Every IPCC default value and uncertainty range pre-loaded in the tool — the energy-equation coefficients, the methane conversion factor Y_m, the maximum methane producing capacity B_o, the manure-system MCF values, and the N_2O emission factors and volatilisation / leaching fractions (EF_3, EF_4, EF_5, FracGASM, FracLEACH) — has been checked line-by-line against the IPCC 2019 Refinement source text (Volume 4, Chapter 10 and Chapter 11). This review confirmed the central values and corrected two pre-loaded uncertainty ranges (B_o and the pasture direct-N_2O factor $EF_{3,\mathrm{PRP}}$) to match the climate-specific ranges in Tables 10.16a and 11.1.

10.5 Reproducibility

Each simulation is driven by an explicit random seed, so any run can be reproduced exactly by re-supplying the same inputs and seed. The convergence guidance in Section “Recommended Iteration Count and Convergence” — re-running with a second seed and checking that the 95% margin of error is stable — lets a user confirm that a reported result is converged rather than an artefact of a particular random draw.

11 Reporting

The results calculated by the tool are designed to align with IPCC uncertainty reporting guidelines (IPCC 2006 Vol 1 Ch 3). The outputs reported by the tool can be used in a hybrid Approach 1 / Approach 2 uncertainty analysis for the whole inventory (IPCC 2006 §3.2.3.3) by entering into Table 3.3 the results reported by the tool for cattle emission sources alongside Approach 1 results for other emission categories. The outputs produced by the tool can also be used as inputs to a full Approach 2 analysis of uncertainty for the whole inventory.

11.1 IPCC Table 3.3 Inputs

In the case of a hybrid Approach 1 / Approach 2, IPCC 2006 Table 3.3 (see also the IPCC 2019 MS Excel tool for Approach 1, and the IPCC 2019 Vol 1 Ch 3 addendum) requires the following inputs to characterise uncertainty of each emission source, each expressed as $\pm x\%$ of the inventory emission estimate:

- Activity-data (population) uncertainty (Column E);
- Emission-factor uncertainty (Column F);
- Combined uncertainty (Column G).

The tool reports all three figures for each cattle emission source via the IPCC Report tab and the Excel / Word downloads. The activity-data vs emission-factor split follows the convention adopted in this tool: AD = animal population (N) only; coefficient (EF) = the IPCC equation parameters that combine into the per-head emission factor (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, Frac_GasMS, Frac_LeachMS, Frac_PRP, EF3_PRP, Frac_GasPRP, Frac_LeachPRP).

11.2 UNFCCC Common Reporting Tables Category Map

The tool can calculate AD, EF and combined uncertainties for one or more cattle categories. While country-specific classifications are allowed for, the UNFCCC Common Reporting Tables (CRTs) indicate reporting of emissions from cattle at the following levels of disaggregation:

- **3.A Enteric fermentation (CH_4)**
  - 3.A.1.a Dairy cattle
  - 3.A.1.b Non-dairy cattle
- **3.B Manure management** (CH_4; direct and indirect N_2O combined)
  - 3.B.1 Cattle (3.B.1.a Dairy, 3.B.1.b Non-dairy)
- **3.D Agricultural soils** (direct and indirect N_2O)
  - 3.D.1.c Urine and dung deposited by grazing animals (direct N_2O)
  - 3.D.2.a Atmospheric deposition (multiple sources, including dung and urine deposited on pasture)
  - 3.D.2.b Nitrogen leaching and run-off (multiple sources, including dung and urine deposited on pasture)

Country-specific options in the CRTs may include reporting of emissions at the level of production systems, regions or cattle sub-categories.

11.3 Disaggregation Levels

To facilitate uncertainty analysis at these different levels of disaggregation, the default outputs in the tool are produced at three disaggregation levels concurrently:

- **Level 1** — disaggregated by cattle type (dairy, non-dairy), emission source (3.A, 3.B, 3.D) and gas (CH_4, N_2O). Available in the IPCC Report tab and the headline summary of the Word / Excel run-summary.
- **Level 2** — Level 1 plus a user-defined second level (production system, region) using the template's `cattle_type` and `region` fields. Available in the per-cattle-type results section of the Word run-summary.
- **Level 3** — Level 2 plus cattle sub-categories (dairy_cow, heifer, bull, etc.). Available in the per-(cattle type × emission source) breakdown of the Word run-summary and the per-sub-category CSV download from the IPCC Report tab.

A user-selectable toggle to choose the reporting disaggregation level for the whole tool is not implemented in this release; the three levels are produced concurrently in their respective output channels. A single-toggle option may be added in a future version after feedback from UNFCCC partners on the right default level.

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